LM1 Forward 远期

定价公式

$FP = S_0 \times (1+rf)^T + \text{Carrying Costs} - \text{Carrying Benefits}$

OC机会成本/时间价值 (Storage Cost) | 金融 < Div, Int. | 实物: Convenience Yield.

$= \begin{cases} ① (S_0 + PVC_0 - PVB_0) \times (1+rf)^T \\ ② S_0 \times (1+rf)^T + FV(\text{Carrying Costs}) - FV(\text{Carrying Benefits}) \end{cases}$

各类Forward定价&估值

or. $= PV_{t,T}[F_t(T) - F_0(T)]$

标的资产		Pricing (标的, 0)	Valuation (合约, t)	
① T-bill	Zero coupon 一年到期	$FP = S_0 \times (1+rf)^T$	$V_t = S_t - \frac{FP}{(1+rf)^{T-t}}$	
② Stock	离散复利	$FP = (S_0 - PVD_0) \times (1+rf)^T$	$V_t = (S_t - PVD_t) - \frac{FP}{(1+rf)^{T-t}}$	只折未来Div 不折过去Div
③ Index	连续复利	$FP = \frac{S_0}{e^{\delta^c \times T}} \times e^{rf^c \times T} = S_0 \times e^{(rf^c - \delta^c) \times T}$	$V_t = \frac{S_t}{e^{\delta^c (T-t)}} - \frac{FP}{e^{rf^c (T-t)}}$	
④ Bond	coupon	$FP = (S_0 - PVC_0) \times (1+rf)^T$	$V_t = (S_t - PVC_t) - \frac{FP}{(1+rf)^{T-t}}$	与Stock相似

剔除股利 $= S_0 - PVD_0$

离散rf → 连续rf^c $1+rf = e^{rf^c} \rightarrow rf^c = \ln(1+rf)$

δ^c: continuously compounded dividend yield (直接给) 股息率

⑤ LIBOR → $\boxed{FRA}$ 概念

远期利率合约.

- long & short: borrow 多头 & lend 空头 (↑赚钱一方, ↑亏钱一方)
- quotation: 1×4 FRA → 1个月后合约到期开始交割(借钱). 4个月后还钱
- 标的利率 LIBOR: single interest (360, USD), floating
- settle in cash 补双方差额

⇓

计算

☆ Settlement $= NP \times \frac{(\text{Floating Rate at Settlement} - \text{Forward Rate})\left[\frac{days}{360}\right]}{1 + \underbrace{\text{Floating Rate at Settlement}}_{\text{(discount rate)}} \times \left(\frac{days}{360}\right)}$ 合约约定 ← 单利折现.

☆ Pricing 确定远期利率. 线性差补法

FR=? 3% 5%

for example: $(1+3\% \times \frac{30}{360}) \times (1+FR \times \frac{90}{360}) - 1 = 1+5\% \times \frac{120}{360} - 1$

Valuation. 2种方法 $\begin{cases} V_t = \text{t时点V相减.} \\ \text{创造相同FRA, 线性差补法算FR. FR之差折现} \end{cases}$

小红书号: 186439052

③ Option Greeks | 5 Greek与option value的正负相关关系

Greeks	Sensitivity factor	Relationship	
		Call option	Put option
Delta	Underlying price	Positive (Delta>0)	Negative (Delta<0)
Gamma	Underlying price	Positive (Gamma>0)	Positive (Gamma>0)
Vega	Volatility	Positive (Vega>0)	Positive (Vega>0)
Rho	Risk-free rate	Positive (Rho>0)	Negative (Rho<0)
Theta	Passage of time t	Closer to maturity → value declines (Theta<0)	Closer to maturity → value declines (Theta<0*)
	Strike Price↑	↓	↑

Delta = 一阶导.斜率
Call Delta [0,1]
Put Delta [-1, 0]
ITM. |Delta|趋近1

rf上升,经济好
特例:
Deep ITM European Put, θ可能>0.

Delta ⟹ Delta Hedge $\frac{1}{\text{Delta}}$
Gamma : 二阶导,凸性. ATM时. Gamma max

No-arbitrage Approach

Hedge Ratio: $h = \frac{\Delta C}{\Delta S} = \frac{C_1^+ - C_1^-}{S_1^+ - S_1^-}$

① Call $h \geq 0$

$h \cdot S_0 - C_0$ 无风险组合 $\begin{cases} h \times S^+ - c^+ \\ h \times S^- - c^- \end{cases}$

$$h \cdot S_0 - C_0 = \frac{hS^+ - c^+}{(1+rf)} = \frac{hS^- - c^-}{(1+rf)}$$

$$C_0 = h \cdot S_0 - PV(h \cdot S^+ - c^+) \quad / \quad = h \cdot S_0 - PV(h \cdot S^- - c^-)$$
$$= h \cdot S_0 + PV(-h \cdot S^+ + c^+) \quad / \quad = h \cdot S_0 + PV(-h \cdot S^- + c^-)$$

② Put $h \leq 0$

$-h \cdot S_0 + P_0$ 无风险组合 $\begin{cases} -h \times S^+ + P^+ \\ -h \times S^- + P^- \end{cases}$

$$P_0 = h \cdot S_0 + PV(-h \cdot S^- + P^-) \quad / \quad = h \cdot S_0 + PV(-h \cdot S^- + P^-)$$

Black Model

着重考公式参数与概念

① Option on Future

$$\begin{cases} C_0 = \frac{FP}{e^{rf \times T}} N(d_1) - \frac{X}{e^{rf \times T}} N(d_2) = e^{-rf \times T}[FP \cdot N(d_1) - X \cdot N(d_2)] \\ P_0 = \frac{X}{e^{rf \times T}} \cdot N(-d_2) - \frac{FP}{e^{rf \times T}} \cdot N(-d_1) = e^{-rf \times T}[X \cdot N(-d_2) - FP \cdot N(-d_1)] \end{cases}$$

② Int Rate Option

$$C_0 = NP \times \underbrace{(accrual\ period)}_{\text{去年化}} \underbrace{e^{-r \times N \times \frac{30}{360}}}_{\text{折现}} [\underbrace{FR_{M \times N}}_{\text{远期rate}} \times N(d_1) - \underbrace{X}_{\text{执行rate.}} \times N(d_2)]$$

$$accrual\ period = \frac{(N-M) \times 30}{360}$$

相似,折现不同.

③ Swaption

Option Underlying: Forward swap rate

✓ Payer swaption r↑ gain 折现系数

$$V_{payer} = NP \times (AP) \times PVA[R_{FIX} \times N(d_1) - R_X \times N(d_2)]$$

✓ Receiver swaption r↓ gain

$$V_{receiver} = NP \times (AP) \times PVA[R_X \times N(-d_2) - R_{FIX} \times N(-d_1)]$$

小红书号: 186439052

LM2 Options Valuation

Put-Call Parity:

$$C_0 + \frac{X}{(1+rf)^T} = S_0 + P_0 \rightarrow \text{call} + \text{bond} = \text{stock} + \text{put}$$

2考点
- ① Replication：等式移动
- ② Arbitrage：buy low, sell high

Binomial Model

每期计算

1. $S_0 \begin{cases} S^+ = S_0 \times u \\ S^- = S_0 \times d \end{cases}$ $u \times d = 1$ → 2. $C^+ = Max[S^+ - X, 0]$, $C^- = max[S^- - X, 0]$ → 3. $\pi u = \frac{1+rf-d}{u-d}$, $\pi d = 1 - \pi u$ → 4. $\frac{C^+ \times \pi u + C^- \times \pi d}{(1+rf)^T}$

拓展

① Two-period

$S_0 \begin{cases} S^+ \begin{cases} S^{++} \rightarrow C^{++} \\ S^{+-} \rightarrow C^{+-} \end{cases} \\ S^- \begin{cases} S^{-+} \rightarrow C^{-+} \\ S^{--} \rightarrow C^{--} \end{cases} \end{cases}$ $\rightarrow C^+ = \frac{C^{++} \times \pi u + C^{+-} \times \pi d}{(1+rf)^T}$, $C^- = \frac{C^{-+} \times \pi u + C^{--} \times \pi d}{(1+rf)^T}$ $\rightarrow C_0 = \frac{C^+ \times \pi u + C^- \times \pi d}{(1+rf)^T}$

② American Option $\begin{cases} C^+, C^{+\prime} \\ C^-, C^{-\prime} \end{cases}$ C提前行权 & C'持有，选高者

美式期权判断是否行权
欧式期权不提前行权

③ Int Rate Binomial Model 利率二叉树
- $\pi u = \pi d = 0.5$, risk neutral probabilities
- $r \neq rf$，结点利率管之后一期

1. $i_0 \begin{cases} i^+ \begin{cases} i^{++} \\ i^{+-} \end{cases} \\ i^- \begin{cases} i^{-+} \\ i^{--} \end{cases} \end{cases}$
2. $C = Max[S-X, 0] \times NP$, $C^{++} / C^{+-} / C^{-+} / C^{--}$
3. $\pi u = \pi d = 50\%$
4. $\frac{C^{++} \times 50\% + C^{+-} \times 50\%}{1+i^+} = C^+$, $\frac{C^{-+} \times 50\% + C^{--} \times 50\%}{1+i^-} = C^-$ $\rightarrow \frac{C^+ \times 50\% + C^- \times 50\%}{1+i^0} = C$

BSM Model

❶ 计算

$$C_0 = S_0 \cdot N(d_1) - \frac{X}{e^{rf \times T}} \cdot N(d_2)$$ 现值×行权概率

$$P_0 = \begin{cases} C_0 + \frac{X}{e^{rf \times T}} - S_0 \\ \frac{X}{e^{rf^c \times T}} \cdot N(-d_2) - S_0 \cdot N(-d_1) \end{cases}$$

$$d_1 = \frac{\ln\left(\frac{S_0}{X}\right) + [R_f^c + (0.5 \times \sigma^2)] \times T}{\sigma \times \sqrt{T}}$$

$$d_2 = d_1 - (\sigma \times \sqrt{T})$$ σ: volatility

2调整

❶ **Options on dividend paying stock**

$$C_0 = [S_0 \times e^{-\delta \times T} \times N(d_1)] - [X \times e^{-R_f^c \times T} \times N(d_2)]$$

$$d_1 = \frac{\ln\left(\frac{S_0 \times e^{-\delta \times T}}{X}\right) + [R_f^c + (0.5 \times \sigma^2)] \times T}{\sigma \times \sqrt{T}} \qquad d_2 = d_1 - (\sigma \times \sqrt{T})$$

❷ **Options on currencies**

$$C_0 = [S_0 \times e^{-r(Foreign) \times T} \times N(d_1)] - [X \times e^{-r(Domestic) \times T} \times N(d_2)]$$

❷ 6 Assumption
- 资产回报率正态分布（=资产价格对数正态分布）
- 资产价格波动率已知且恒定
- 无风险利率已知且恒定
- 市场完全有效
- 标的资产未来不产生现金流（若有CF，则 S^c: $S_0 \Rightarrow \frac{S_0}{e^{\delta^c \times T}}$）
- 只用于欧式期权

Bywwcnll
小红书号：186439052

LM1 Swap

一定注意 c 是 period 的，题目给 fixed rate 是一年的

Interest Rate Swap

同种货币固 vs 浮

Ⓐ fixed pay - floating side　0 时点交易双方无交易
Ⓑ floating pay - fixed side.　交易双方差额交割.
即 A 发行 floating bond，买入 fixed bond. B 反之

PV=1　c　c　c　c+1　本金=1
0　90　180　270　360　c=2%

❶ Pricing 确定固率
- ① $r_{90}, r_{180}, r_{270}, r_{360} \Rightarrow B_1, B_2, B_3, B_4$ 算折现因子
- ② $\frac{1-B_4}{B_1+B_2+B_3+B_4} = c$ (periodic) 求 c.
- ③ $c \times 4$ = swap rate. 年化

❷ Valuation
- $r_{60}^{30}, r_{60}^{120}, r_{60}^{210}, r_{60}^{300} \Rightarrow B_1', B_2', B_3', B_4'$
- $V(\text{fixed}) = c \times (B_1' + B_2' + B_3' + B_4') \times 1 + B_4'$
- $V(\text{floating}) = (1+r_{90}) \times B_1'$

二者相减 × NP

60　c　c　c　c+1
0　90　180　270　360
reset day：PV=面值

之后 reset day PV=面值
∴ 未来 CF 折到 90 天，为面值！

Currency Swap

❶ Difference：交换 NP / 4 种 swap / Exchange Rate

❷ Pricing：同 interest rate swap，算 2 种币种 fixed rate.

❸ Valuation
- ① B_1' B_2' B_3' B_4'（2 组：2 种币种）.
- ② \$V fixed / \$V floating / £V fixed / £V floating 同 IR swap
- ③ $\$V_t - \boxed{£V_t \text{ 换成等额 NP} \times \text{FX rate.}} = V_t$

Equity Swap

❶ Pricing：同 IR swap

❷ Valuation
- V fixed & V floating 同 IR swap
- V_{Equity} = NP × (1+ 波动率)

小红书号：186439052

LM1 T-bond Futures. 国债期货

特殊的远期合约. 每日归零, 无需估值, 定价 = Forward

概念

❶ CTD option $\Rightarrow$ $FP_{标} \times CF_A = FP_A$

标准化债券 ↓ ($FP_{标}$)　　实际交割债券. 交割价 ↓ (FP_A)

Short方选择 cheapest-to-deliver bond　　conversion factor 转换因子

交割价与市价比, 亏损最少的 Bond 用来交割 "CTD"

❷ 报价: $95 + \frac{1}{32}$

Arbitrage Profits

Full Price 全价. 交割价
Clean Price 净价. 报价.
$>$ Full = Clean + Accrued Interest.

无套利 $\begin{cases} V_1 = FP_{标}^{clean} \times CF_A + AI_T \\ V_2 = (S_0^{clean} + AI_0) \times (1+R_f)^T - FVC_T \\ |V_1 - V_2|_T \xrightarrow{折现} 0 \end{cases}$ 从而求 $FP_{标}^{clean}$

AI_T: Accrued Interest

V_1: 买入A的期货

V_2: 0时点买入A bond 现货

T时点持有A的2种方式

Pricing

$$\text{Quoted Future Price}^{clean} = \left[\left(\underbrace{S_0^{clean} + AI_0}_{S_0^{full}} \right) \times (1+R_f)^T - \underset{①}{FVC} - \underset{②}{AI_T} \right] \times \underset{③}{\frac{1}{CF}}$$

三步调整

$FP = (S_0 - PVC_0) \times (1+r_f)^T$　　由远期的2步推理

① $= (S_0^{clean} + AI_0) \times (1+r_f)^T - FVC$

② $= [(S_0^{clean} + AI_0) \times (1+r_f)^T - FVC - AI_T] \times \frac{1}{CF}$

变为clean price. 变为标准 bond

Bywwcnll
小红书号: 186439052